U.S. Flight Delay Propagation

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Introduction

Current flight delay information is presented through airline reports, airport statistics, formulas, or basic dashboards. Existing research analyzes delays and how they propagate mathematically, but offers limited interactive exploration.

Our project addresses this gap through **interactive visualizations** that help users understand the structure of the U.S. flight network and explore how delays propagate nationwide.

Visualization Approach

Our system innovates by identifying network structure, hub importance, and cascading disruptions through interactive exploration that users can manipulate.

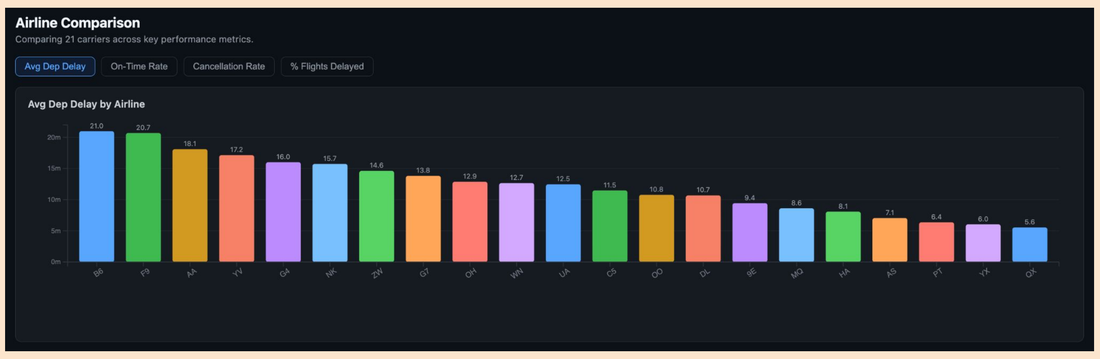
These views include the following: a **Network Map** mapping 377 U.S. airports, an **Airport Detail View** highlighting any airport with calculated statistics, a **Delay Propagation View (Novel Innovation)**, animating a radial tree visualization that shows how delays spread outward from a airport, and a **Airline Comparison View** exploring delay trends across airlines.

Algorithm Approach

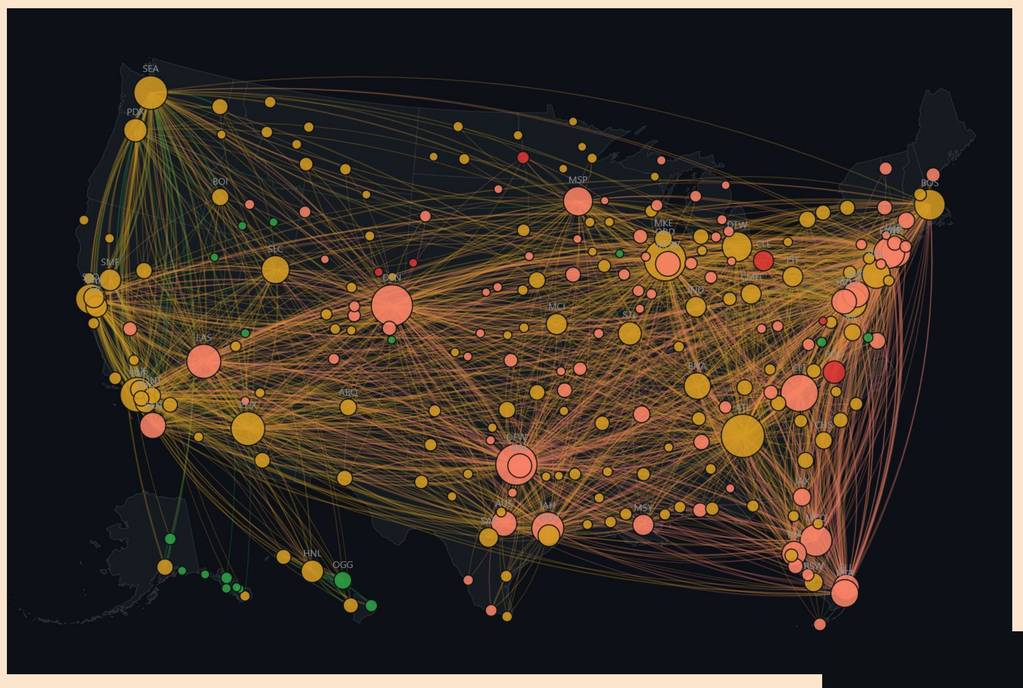
We cleaned U.S. domestic flight records to **25.9 million valid flights**. Aggregation was performed in **DuckDB**, generating tables for **377 airports, 7,565 routes**, hourly delays, airline performance, and monthly trends.

A key challenge was identifying when delays at one airport likely caused delays elsewhere using only public data. Our method uses a **novel temporal correlation algorithm** on open-source data that found **9.66 million raw propagation events**.

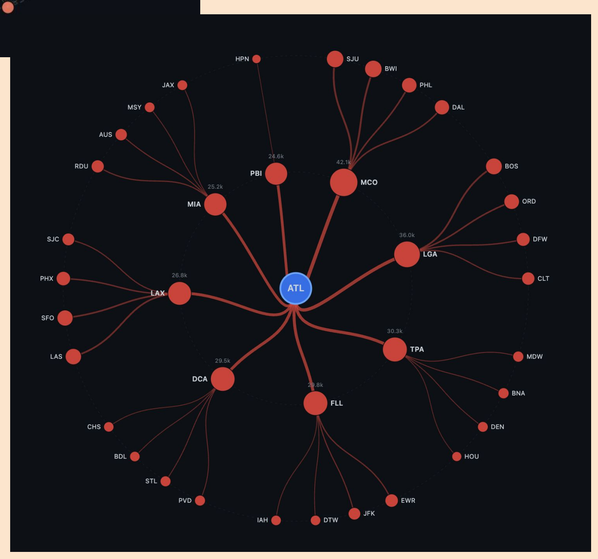
The intuition behind our approach is that aircraft routing creates a dependency chain (a late inbound plane physically cannot depart on time). Through our algorithm, we are able to approximate the conditions that allow for one delay to cause another. We flag propagation events when consecutive flights are 15+ minutes late, and departures are between 30 and 90 minutes apart.



The **Airport Details** and **Airline Comparison** views (above) show advanced details for each airport, as well as comparison between different airlines on the basis of metrics such as average departure delay and cancellation rate. For example, the first view allows us to see the delay by hour of day for ATL (as an example) as well as noticing how ATL only has a 62% on-time rate.



The **Network Map** view (left) showing the **377 U.S. airports.** Darker red nodes represent airports with higher average departure delay times, while node size represents flight volume. DEN stands out for being a major east-west hub with moderate delays.



The novel **Delay Propagation View** (right) showing the **inferred delay propagations** from ATL. From ATL, the next radial ring of airports represents the next set of airports where a delay at ATL will propagate to, and so on and so forth through concentric rings. Users can interactively add/remove outer rings or adjust a threshold to filter weaker propagation links. This novel re-rooting interaction lets users follow delay chains dynamically, which static dashboards cannot do.

Experiments and Results

Our evaluation focuses on **three questions:** whether users can successfully navigate the system **without prior training**, whether the visualizations **clearly communicate** delay and propagation patterns, and whether the algorithm's outputs **align with real-world** aviation behavior.

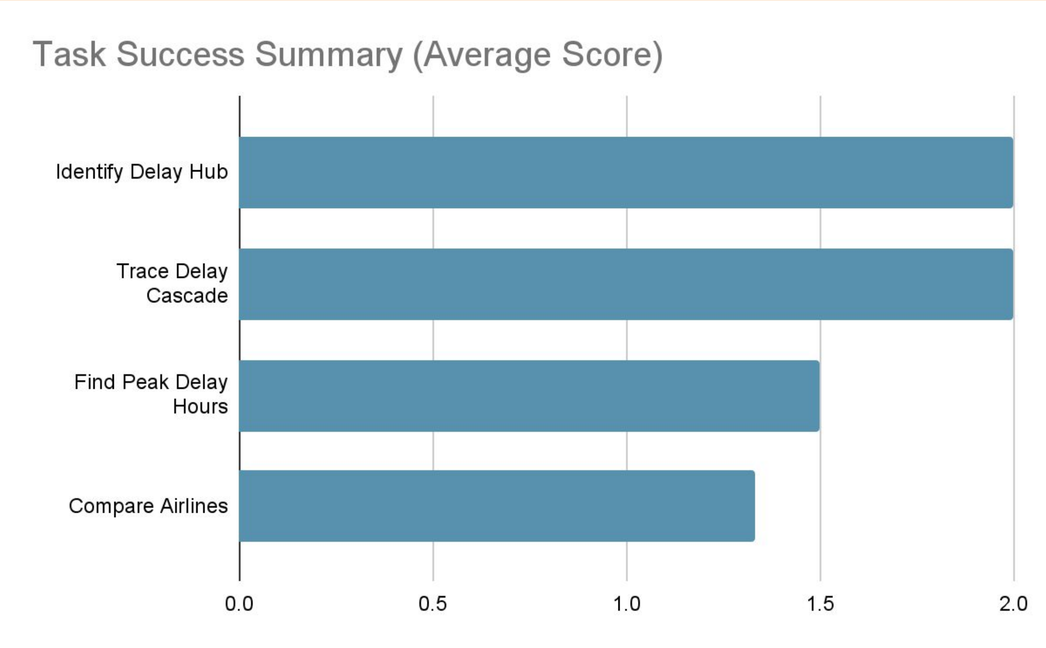
User Study Key Findings

- Users completed major analytical tasks successfully.
- Interface rated highly for clarity and navigation.
- Main requested improvements were additional filters and reducing clutter.

Network Analysis Results

1. ATL: high degree centrality (0.86) and over 1.2 million incoming flights, confirming its central role in the network.
2. DEN: strong betweenness centrality (0.164), showing its importance as a bridge between east/west U.S. routes.
3. DFW: high PageRank (0.0412), indicating strong connections to other major hubs.

These results suggest our methods capture meaningful structure in both the flight network and delay propagation system.



Data

Our project uses publicly available U.S. domestic flight performance data **downloaded** from the **Bureau of Transportation Statistics (BTS)**. The dataset contains detailed records for scheduled domestic flights (2022 - 2025), including airports, airlines, departure times, arrival times, cancellations, and delay durations.

In total, the raw dataset contains **26.3 million flight records** taking up **1.61 GB**, making it both large-scale and **temporal**. We downloaded our dataset from Row Zero.